

tests/regression/ assert_egress_ip_host_unknown_test.sh — operator guide

Revision: 2 **Last modified:** 2026-07-01T12:00:00Z **Status:**
Active standing regression guard (§11.4.135) for finding F7 (+ F-1 hardening).

Companion (§11.4.18) to the in-source documentation block at the top of the script. Sibling of docs/scripts/external_egress_verdict_test.md.

Overview

A standing regression guard that proves tests/lib/evidence.sh's `assert_egress_ip()` never **fake-PASSes** the VPN-routing proof when the host's **real IP is not a trustworthy public IP** — unknown, empty, non-IP garbage (a captive-portal / rate-limit HTML body a `curl -s 200` can echo), or a non-public sentinel like `0.0.0.0` / `127.x` (**F-1 hardening**).

The proof (design §15) has **two independent halves**:

1. `egress == expected_exit` — traffic left via the expected VPN tunnel exit; and
2. `egress != host_real` — traffic did **not** simply leave the host's own uplink (an egress that equals the host's real IP means **no VPN routing happened** — the §15 bluff).

Pre-fix, when a caller's IP-echo fallback fired — `curl ifconfig.me` || `echo "unknown"` in `verify-proxy.sh` / `final-verify.sh` / `comprehensive-test.sh`, or || `true` (empty) in `real_vpn_egress_proof.sh` — `host_real` became the literal `"unknown"` / `""`. Comparing the observed egress against `"unknown"/""` trivially satisfies `"different"`, so the `egress != host` half **silently collapsed**: a genuine `egress == host` (no-VPN) case could still PASS. That is the §11.4.68 **fail-open** — the anti-VPN-bluff check losing half its assertion the moment the host IP is unknown.

The fix: when `host_real` is unknown/empty the `!=host` half is **unverifiable**, so the call is classified as follows (never a fail-open PASS):

host_real	observed vs expected_exit	verdict	rc	meaning
known	== exit, != host	PASS	0	genuine proof — both halves verified
known	== host	FAIL	1	the §15 bluff — no VPN routing
known	!= exit	FAIL	1	wrong exit
unknown / "" / garbage / sentinel	!= exit	FAIL	1	wrong exit is a provable defect regardless of host
unknown / "" / garbage / sentinel	== exit	OPERATOR- BLOCKED	2	!=host half unverifiable — never PASS; §11.4.69 reason network_unreachable_external; rerun once the host public IP is determinable

F-1: host_real is validated to be **IP-shaped** (IPv4 with 0-255 octets, or an IPv6 hex:colon form) via _evidence_ip_shaped rather than deny-listing the two literal sentinels ""/"unknown". A non-empty, non-"unknown" **garbage** value (HTML body) or a non-public sentinel (0.0.0.0, 127.x, ::, ::1) is exactly as unverifiable as an empty one and takes the same exit-2 branch — closing the residual window where a garbage host_real could re-collapse the !=host half.

Return 2 (OPERATOR-BLOCKED) rather than a return-0 SKIP is deliberate: for a sink-unreachable condition §11.4.68 mandates **exit-2 OPERATOR-BLOCKED, never a fail-open SKIP-as-PASS**. Existing if assert_egress_ip ...; then test_pass; else test_fail; fi callers route the non-zero into their FAIL/block branch — safe (the old behaviour would have fake-PASSED).

It does **not** grep the source. GREEN sources the **real** shipped assert_egress_ip and drives it through the committed EVIDENCE_OBSERVED_IP_FILE fixtures — deterministic, no network.

Prerequisites

- POSIX sh, awk, tr (via sourced evidence.sh). No network, no privileges.
- Committed fixtures tests/lib/fixtures/egress_observed_vpn.ip (185.65.135.70) and egress_observed_host.ip (203.0.113.45).

Usage examples

```
# GREEN guard (default) – fail-open closed, provable defect FAILs,  
  genuine PASS kept:  
tests/regression/assert_egress_ip_host_unknown_test.sh  
  # exit 0 = PASS
```

```
# RED reproduce – the pre-fix replica fake-PASSes the hidden §15  
  bluff:  
RED_MODE=1 tests/regression/assert_egress_ip_host_unknown_test.sh  
  # exit 0 = defect reproduced
```

§11.4.115 RED_MODE polarity

RED_MODE	What it runs	PASS means
0 (default)	the REAL shipped assert_egress_ip (sourced from evidence.sh)	host unknown/" " + egress==exit → 2; host unknown + egress==host → 2; host unknown + wrong exit → 1; host known + egress==exit && !=host → 0 (GREEN guard)
1	a faithful PRE-FIX replica (3-check logic, no host- unknown guard)	egress==host==expected with host "unknown" → rc 0 PASS → the §11.4.68 fail-open reproduces

A RED_MODE=1 run that cannot reproduce is a finding per §11.4.7, not a pass.

Edge cases

- **Fix reverted / guard neutered** (the host_real unknown/empty branch removed or made unreachable) → the GREEN guard reports REGRESSION: unknown/empty host no longer refused ... and exits 1 (the §1.1 paired-mutation behaviour, verified against the real function).
- **Wrong-exit under unknown host** stays a FAIL(1) — an egress that is definitively not the expected VPN exit is a provable defect independent of host knowledge, so the guard must **not** let the unknown-host branch swallow it into an OPERATOR-BLOCKED.
- **Genuine PASS preserved** — a fully-known, correctly-routed case (egress==exit && !=host) must still return 0; the guard asserts this so the fix cannot over-block.

Internal behaviour

- `#!/bin/sh, set -eu; POSIX-only, sh -n / bash -n parse-clean` (§11.4.67).
- GREEN sources `tests/lib/evidence.sh` and drives the real `assert_egress_ip` via the `EVIDENCE_OBSERVED_IP_FILE` unit-test seam with the committed fixtures.
- RED runs a self-contained pre-fix replica (no network, no seam).
- Writes one evidence file per run under `qa-results/regression/assert_egress_ip_host_unknown/` (gitignored).

Related

- Fix site: `tests/lib/evidence.sh — assert_egress_ip()` host-unknown guard.
- Unit-layer cases: `tests/lib/evidence_selftest.sh` — the F7 block (`run_case 2 ... host UNKNOWN/EMPTY ...`, `run_case 1 ... WRONG exit ...`).
- Callers protected: `tests/verify-proxy.sh`, `tests/final-verify.sh`, `tests/comprehensive-test.sh`, `tests/egress_proof/real_vpn_egress_proof.sh`, `tests/dynamic/analyzers/egress_neq_host_analyzer.sh`.
- `tests/run-tests.sh` — registers this guard (GREEN + RED_MODE=1) in the standing `test_regression_guards()` suite as `BUGFIX-0018 assert-egress ...` (§11.4.135).
- `docs/issues/fixed/BUGFIXES.md` — BUGFIX-0018 (F7 + F-1).

Last verified

2026-07-01 — `sh -n / bash -n parse-clean`; GREEN asserts 2/2/2/1/0 for the five canonical cases **plus** the two F-1 cases (0.0.0.0 sentinel + garbage-HTML hidden bluff → 2/2) (exit 0); RED_MODE=1 reproduces the fail-open (exit 0); paired §1.1 mutation (`return 2` → `return 0` in the unverifiable-host branch) makes GREEN FAIL (exit 1 — "fail-open re-opened"), then restores byte-identical (md5 976cb15... before == after); full `evidence_selftest.sh` = 45 pass / 0 fail; registered in the standing `run-tests.sh` suite (full run: 59 run / 53 pass / 6 skip / 0 fail).